outer product : xy^T

$x \in \mathbb{R}^p \quad y \in \mathbb{R}^q$

$\hookrightarrow \; xy^T \in \mathbb{R}^{p \times q}$

inner product : $x^T y$

$= x_1 y_1 + x_2 y_2 + \cdots + x_n y_n = y^T x$

same as dot product

I is unit matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$AI = IA = A$

$AB = I$

$\uparrow$ inverse matrix of A

$B = A^{-1}$

$(A^T)^{-1} = (A^{-1})^T$

* $A^{-1}y = A^{-1}Ax = Ix = x$

Special matrices

① square matrix , $m = n$

② Identity matrix $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

③ if $m=n$ and has full rank for matrix A, then A is invertible, therefore null space is empty.

if not full rank, then A is not invertible, therefore nullspace is not empty

④ for invertible matrix:

$$A^{-1}A = AA^{-1} = I$$

⑤ $n \times n$ matrix where $A^T = A$ this is called <u>symmetric</u>

⑥ if $A^T = A$ and if $x^T A x > 0$ for every $x \neq 0$, this is called <u>symmetric positive definite</u> (SPD)

⑦ if $A^T = -A$ is called <u>skew symmetric</u>

⑧ $n \times n$ matrix where

$$A^T A = A A^T = \underline{I}$$

is called <u>orthogonal</u>.

⑨ $n \times n$ matrix where $a_{ij} = 0$ for $i \neq j$ is called <u>diagonal</u>

(10) matrix is sparse if most elements in the matrix are zero. A diagonal matrix is a sparse matrix

(11) lower triangular matrix:

where $a_{ij} = 0$ for $i > j$

upper triangular matrix:

where $a_{ij} = 0$ for $i < j$

(12) determinant of square matrix:

$$\det(A) = a_{11}a_{22} - a_{21}a_{12}$$

some properties:

* $\det(AB) = \det(A)\det(B)$
* $\det(A^T) = \det(A)$
* $\det(\alpha A) = \alpha^n \det(A)$
* if A is invertible,

then $\det(A) \neq 0$

and $\det(A^{-1}) = \dfrac{1}{\det(A)}$

(B) if we have SPD matrix P,

then $P^{1/2}$ exists, where:

$$\det(P) = \det(P^{1/2} P^{T/2})$$

$$= \det(P^{1/2})^2 \quad \text{so that}$$

$$\det(P^{1/2}) = \det(P)^{1/2}$$

Overdetermined least squares

* collection of observations, aka our data set : $y \in \mathbb{R}^m$

this is a column vector

* connected to underlying physical model : $x \in \mathbb{R}^n$

* $A \in \mathbb{R}^{m \times n}$

we can make prediction that $y = Ax$, which is only true if :

① $m=n$ ② A is full rank

example :

$$\underline{x_1} + 2\underline{x_2} - 4\underline{x_3} = y_1$$
$$2x_1 - 5x_2 + x_3 = y_2$$
$$4x_1 + 3x_2 - 3x_3 = y_3$$

I can solve this exactly.

However, if $m > n$, now we can say:

$$y \approx Ax$$

this is overdetermined

define a residual, r:

$$r = y - Ax$$

we want solution that minimizes r. In simple LS, this means minimize 2-norm of r.

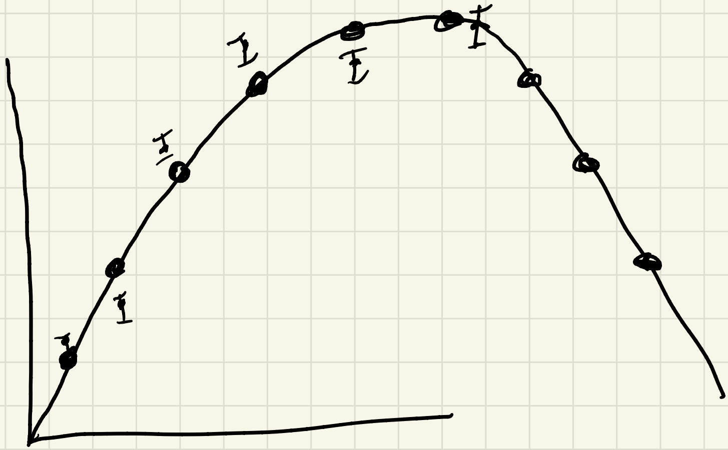
$$\hookrightarrow f(x) = \|r\|^2 = r^T r$$
$$= (y - Ax)^T (y - Ax)$$

<u>normal equations</u>

$$A^T A x = A^T y$$

$$\hookrightarrow x = (A^T A)^{-1} A^T y$$

$$0 = A^T (Ax - y) = A^T r = \begin{bmatrix} a_1^T r \\ a_2^T r \\ \vdots \\ a_n^T r \end{bmatrix}$$



$Gm = d$

$\Rightarrow Ax = y$

math model:

$$y(t) = m_1 + m_2 t - \frac{1}{2} m_3 t^2$$

$$\begin{bmatrix} 1 & t_1 & -\frac{1}{2}t_1^2 \\ 1 & t_2 & -\frac{1}{2}t_2^2 \\ \vdots & \vdots & \\ 1 & t_m & -\frac{1}{2}t_m^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_m \end{bmatrix}$$

$m > n$